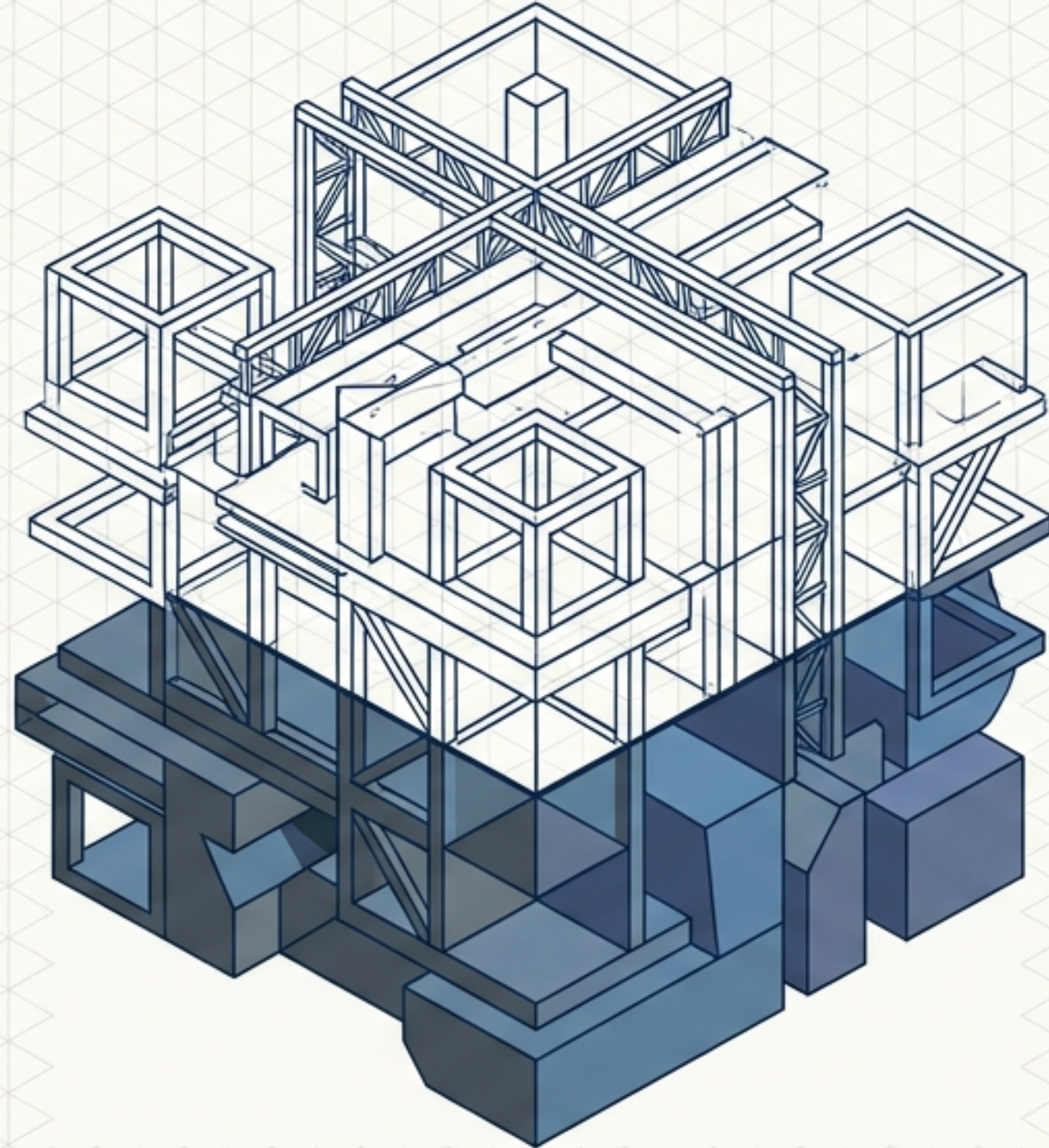
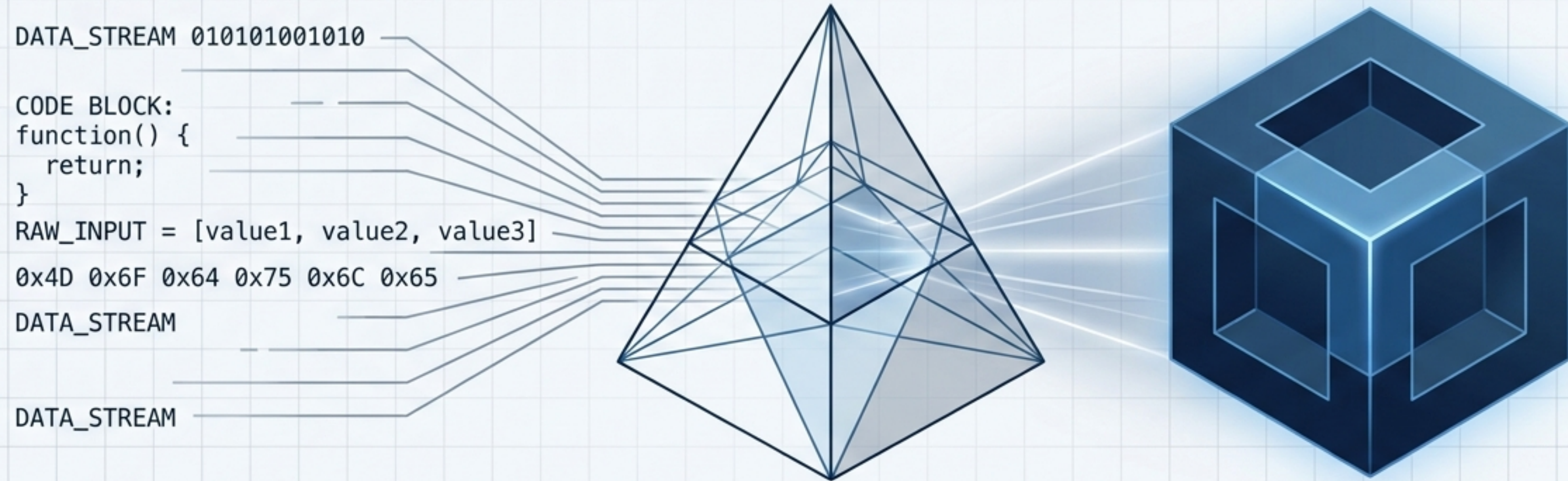


The Architecture of Automation

A Comprehensive Guide to Modern CASE Tools



Defining the Concept: What is CASE?



Software Engineering Methods + Automation = High Quality, Zero Errors, Seamless Maintenance.

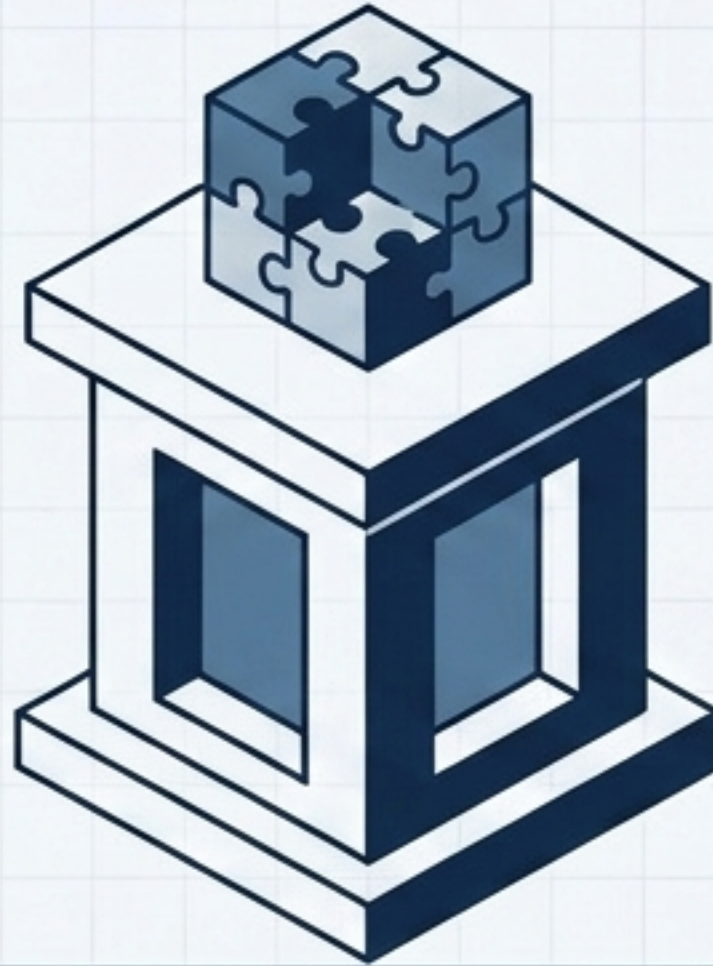
Computer-Aided Software Engineering (CASE) tools are the instruments used to automate the design and development of information systems.

The DNA of CASE: Three Core Characteristics



Powerful Graphical User Interface

Optimized for user interaction and system description.



Component Integration

Seamless workflow across distinct tools.



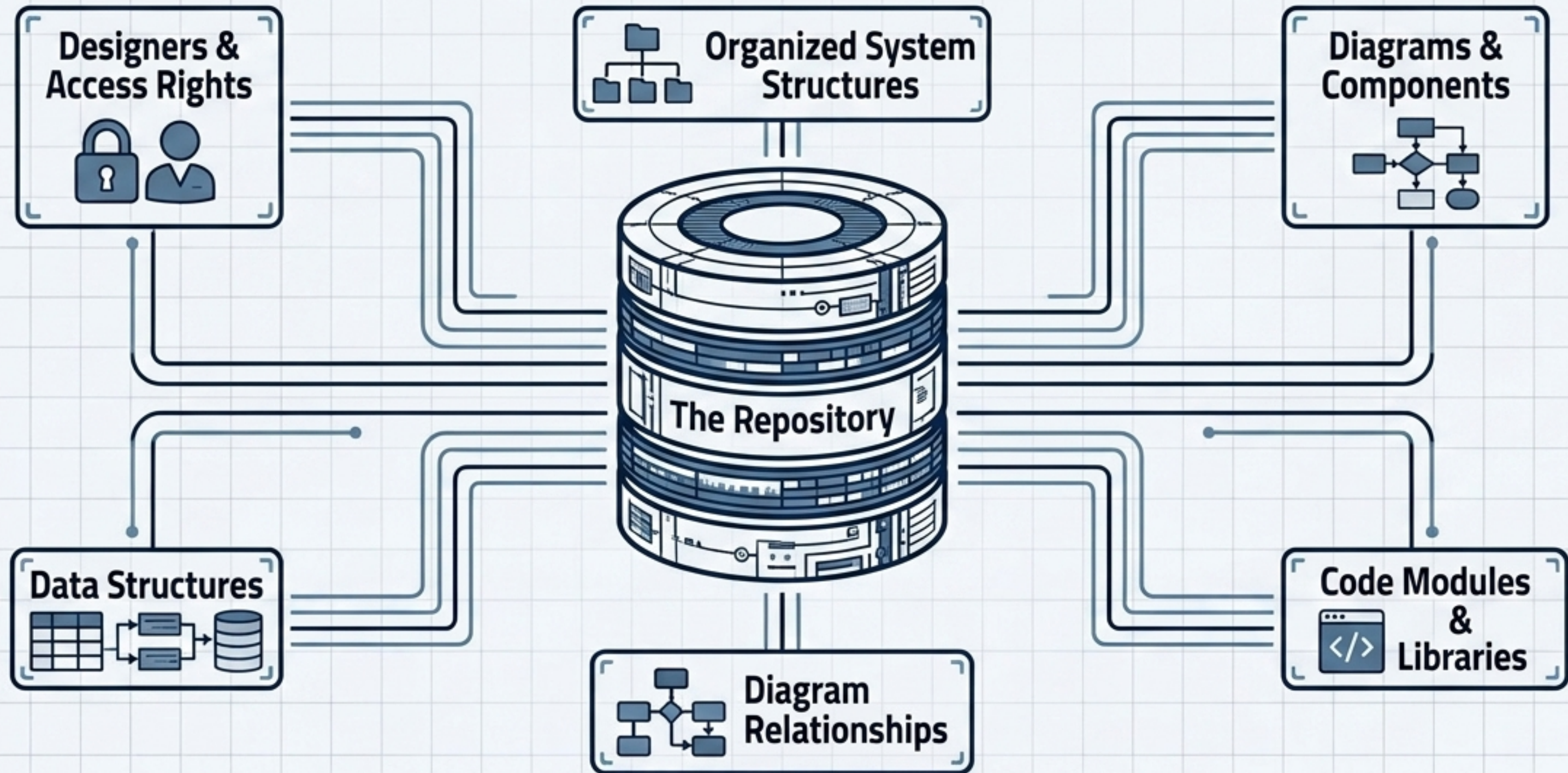
Centralized Repository

The single source of project truth.

A true CASE tool must possess all three characteristics.

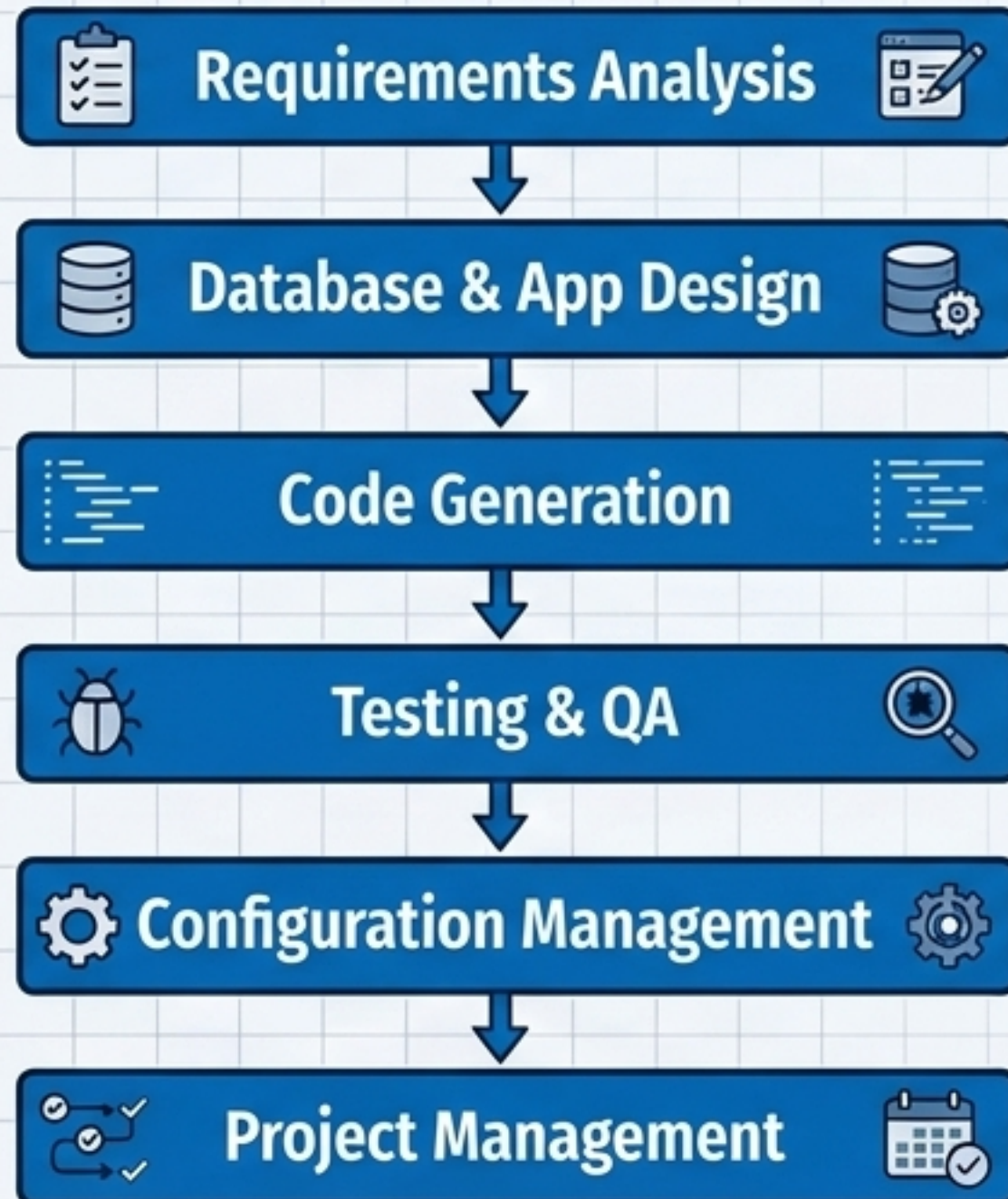
The Engine: The Centralized Repository

The specialized database that captures the system's exact state at any given moment.



Lifecycle Automation & ROI

The Process



The Impact

**Optimized
System
Structure**

**Significantly
Lower
Development
Costs**

**Higher
System
Efficiency**

**Reduced
Probability
of Design
Errors**

The CASE Taxonomy: Navigating the Market

By Methodology

Functional /
Structural

Object-Oriented

Complex

By Functional Type

Analysis

e.g., BpWin

Database Design

e.g., ERwin, Oracle Designer

App Development

e.g., Delphi, Oracle Developer

Reengineering

e.g., Rational Rose

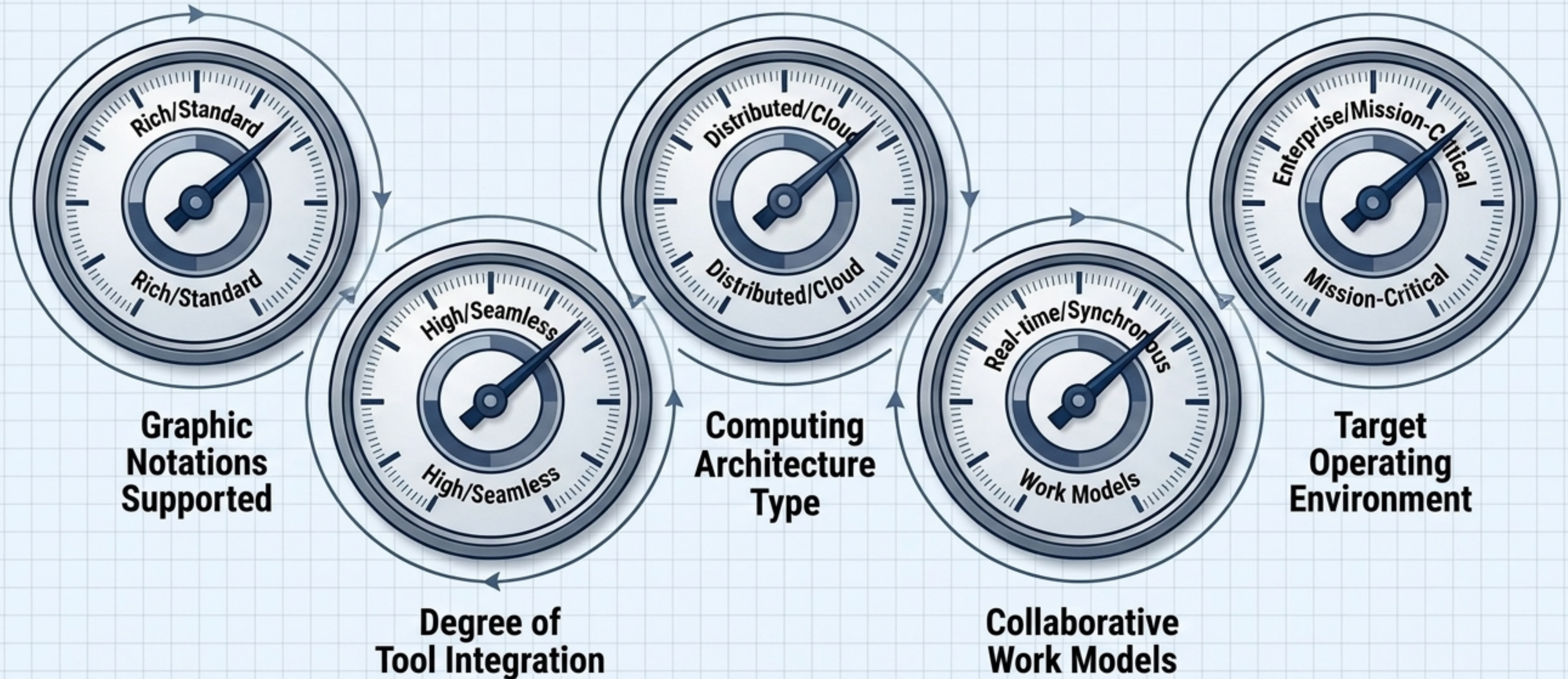
By Integration Category

Local
(Autonomous tasks)

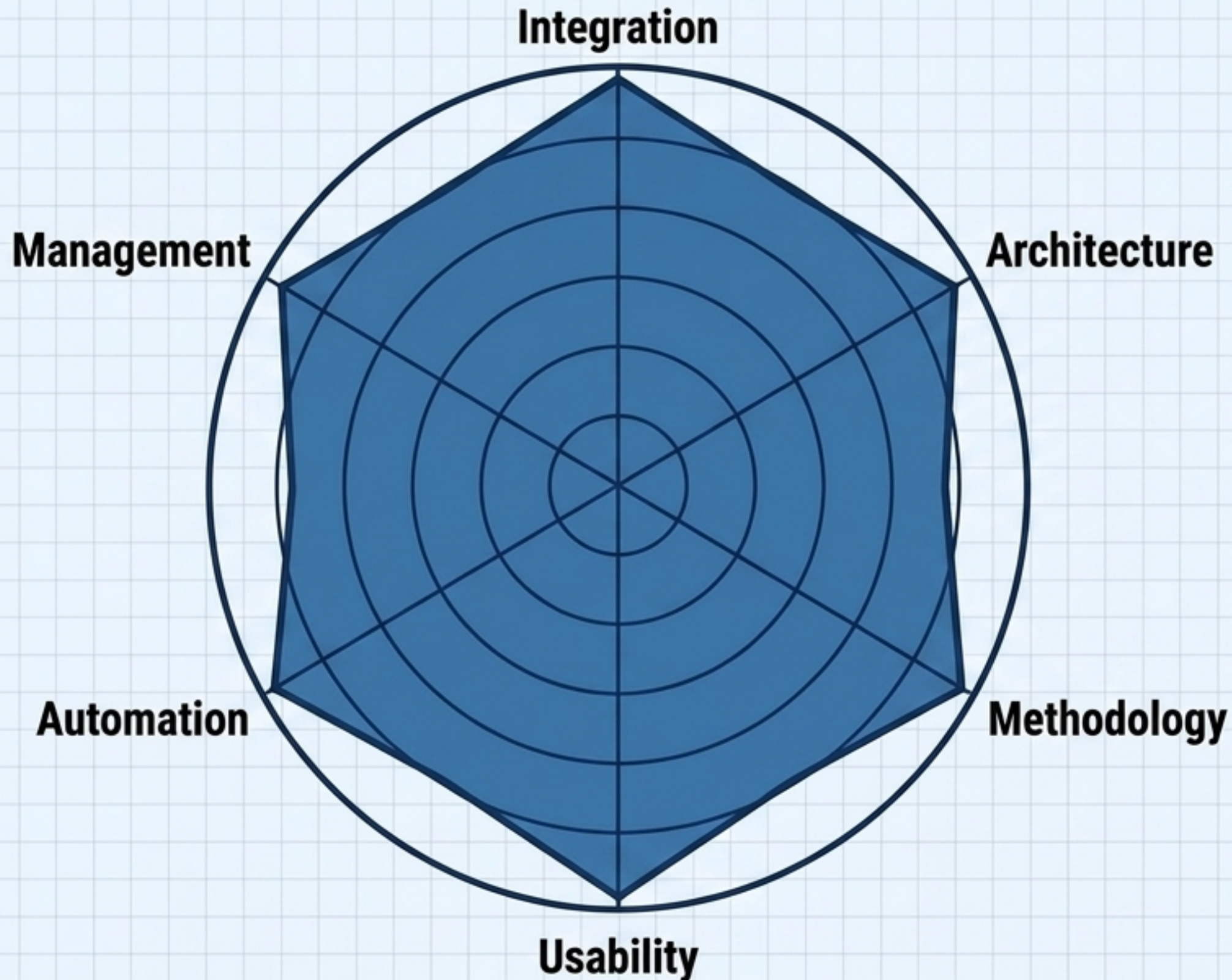
Partially Integrated
(Spanning multiple
phases)

Fully Integrated
(Entire lifecycle tied to a
single repository)

Configuring the Environment: Technical Differentiators



The Selection Diagnostic Matrix



Presence of
Database or
Data Dictionary



Required
Methodology
Support



External
Interfaces & Tool
Compatibility



High-Quality
Graphical
Support



Robust Import
& Export
Capabilities



Automatic
Code
Generation



Open
Architecture
Standards



Built-in
Project
Management

The Lingua Franca of Modeling: UML

U

Provide developers with a unified visual modeling language.

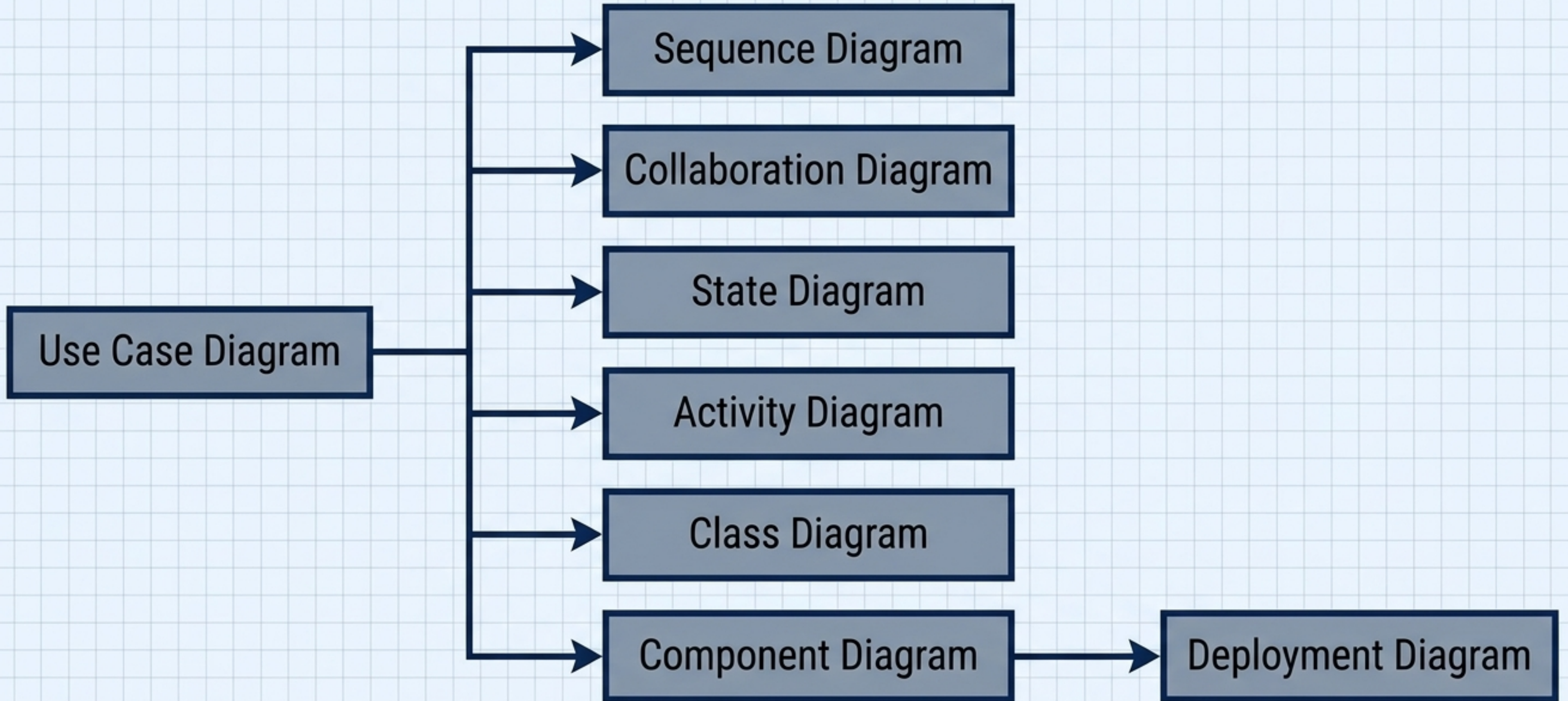
M

Provide mechanisms for language extensibility and specialization.

L

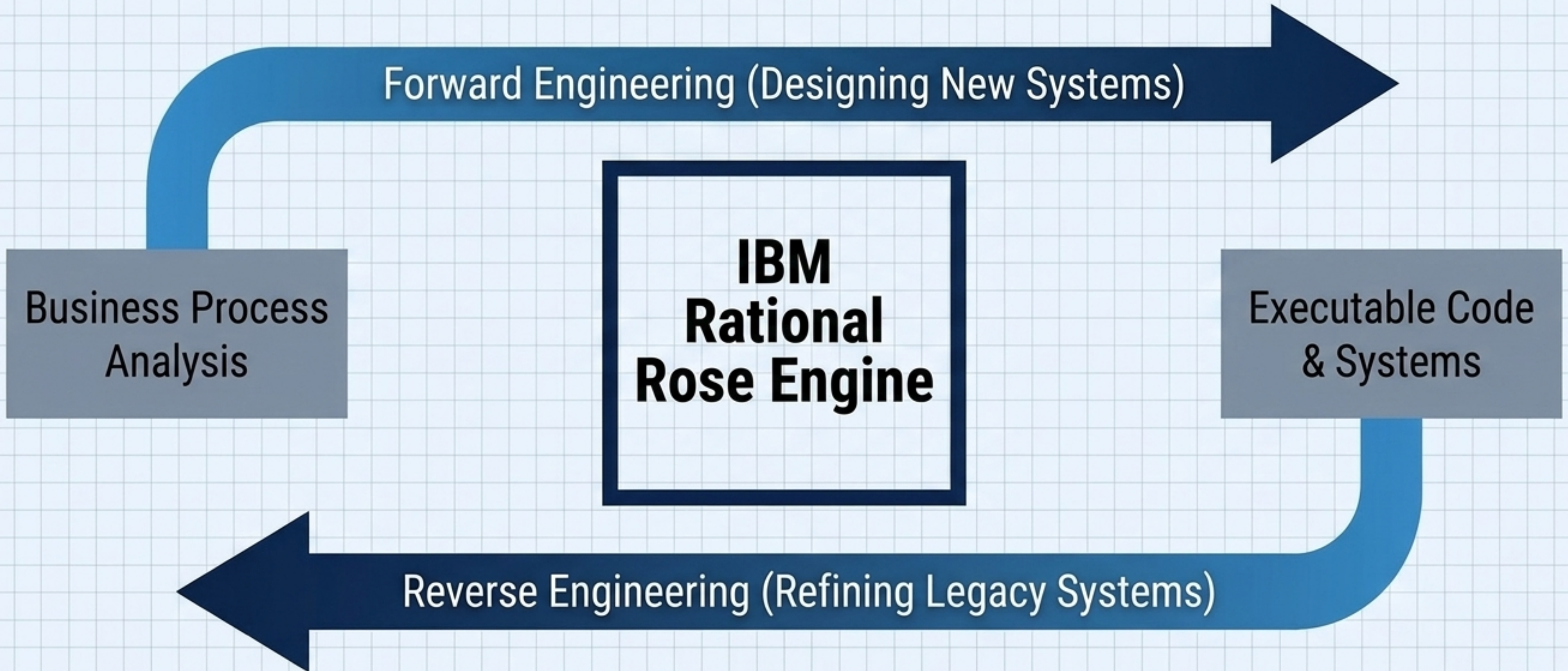
Ensure independence from specific programming languages and development processes.

The UML Ecosystem Map



Spotlight: IBM Rational Rose

A comprehensive tool spanning the entire software lifecycle.



Rational Rose Technical Capabilities

Supported Languages

	ADA
	Java
	C
	C++
	Basic

Supported Technologies

	COM
	DDL
	XML

Database Schema Generation

	Oracle
	SQL

Rational Rose: Edition Matrix

Edition	Business Analysis	Forward / Reverse Eng.	100% Executable Code	Primary Focus
Modeler	✓	—	—	System Design
Professional	✓	Language Dependent	Framework Only	Standard Dev
RealTime	✓	C / C++ Only	Yes (Real-Time)	Auto-Compilation
Enterprise	✓	Full Spectrum	—	Enterprise Lifecycle
DataModeler	—	—	—	Database Schema Design

The Engineering Discipline Realized



By combining centralized repositories, standardized modeling (UML), and full-lifecycle tools (Rational Rose), CASE methodology transforms software development from an unpredictable art into a precise, automated, and zero-error engineering discipline.